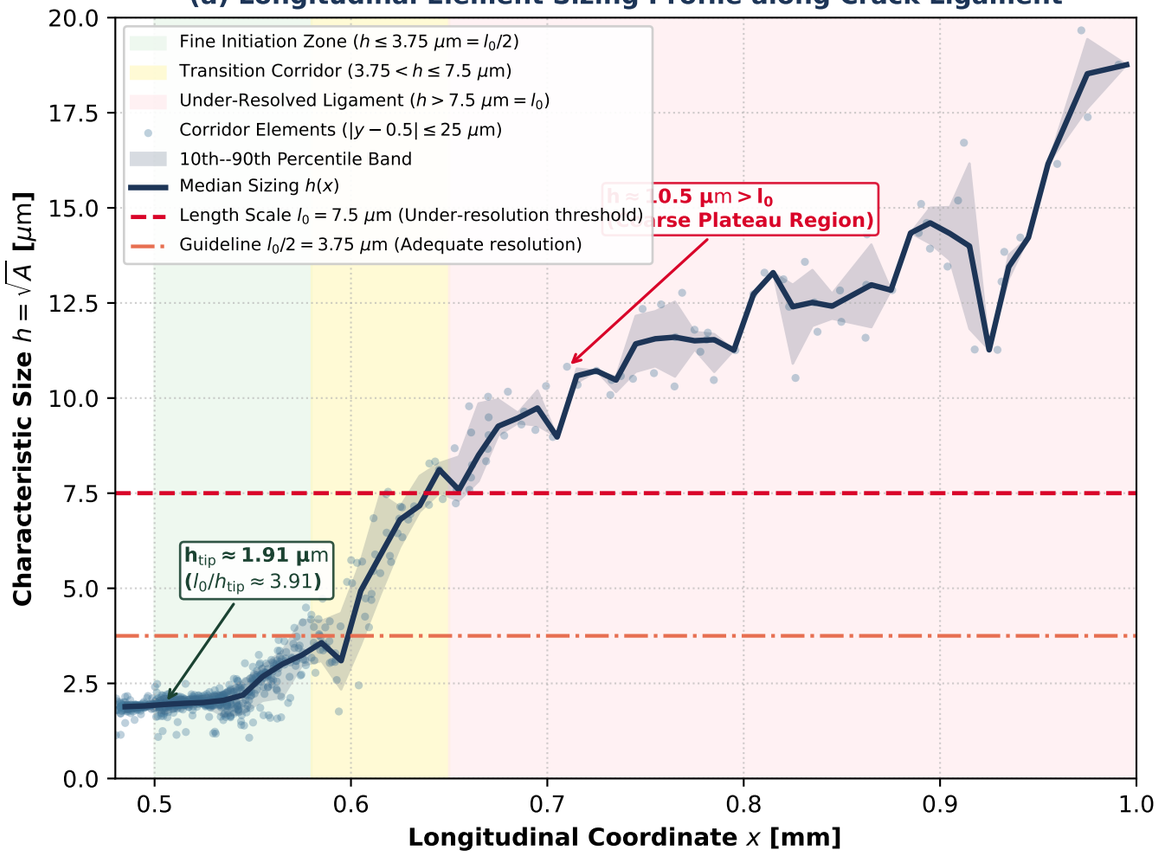
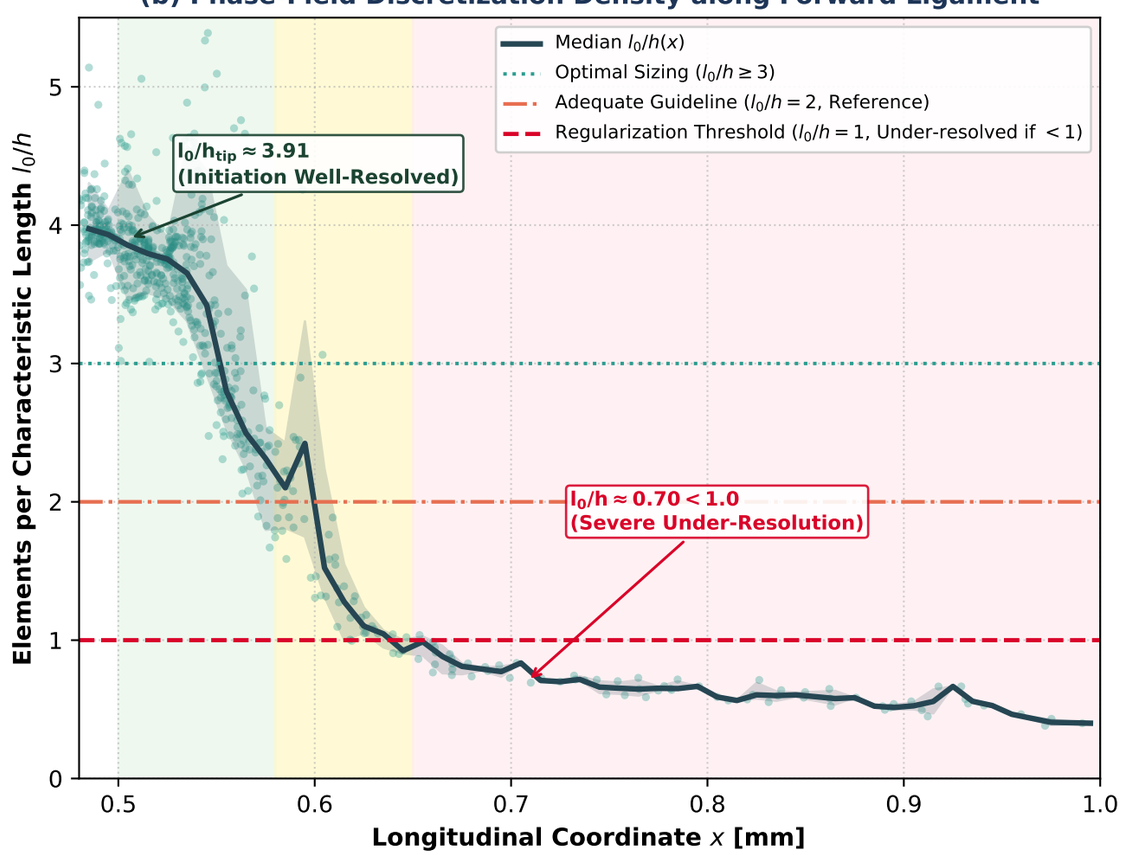
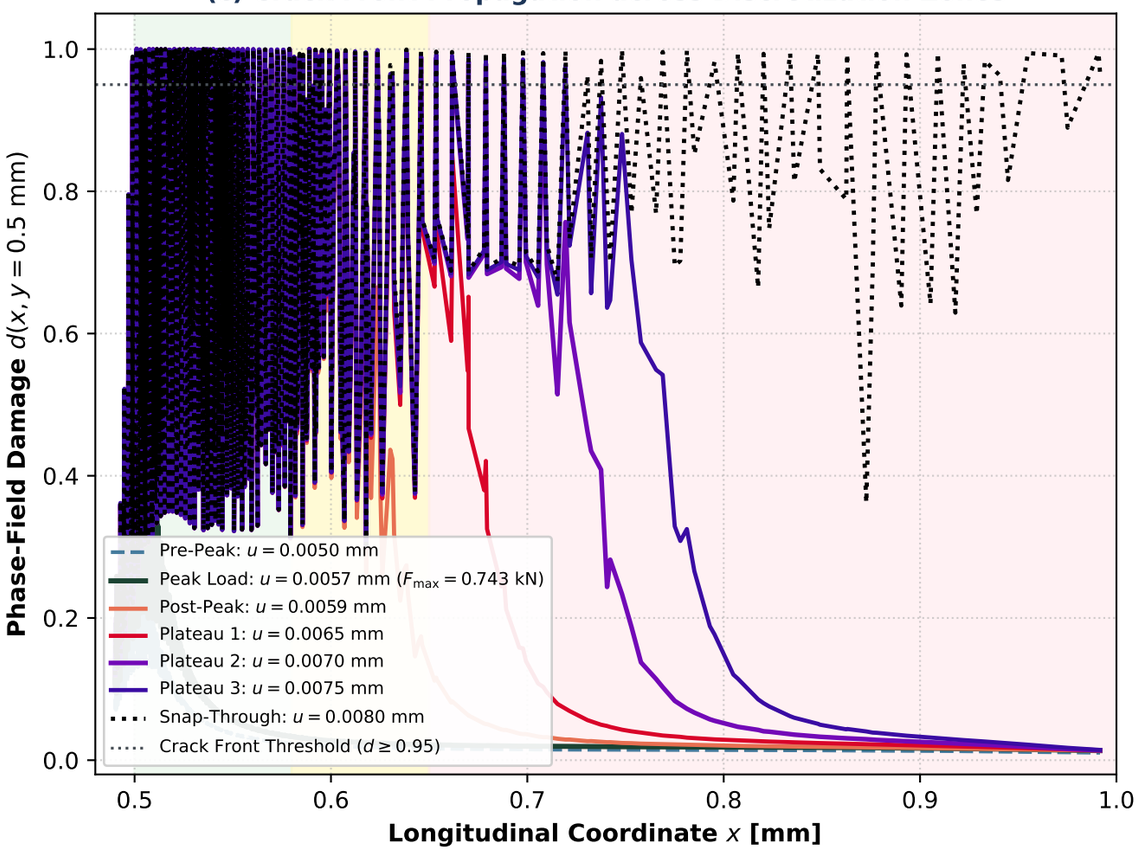
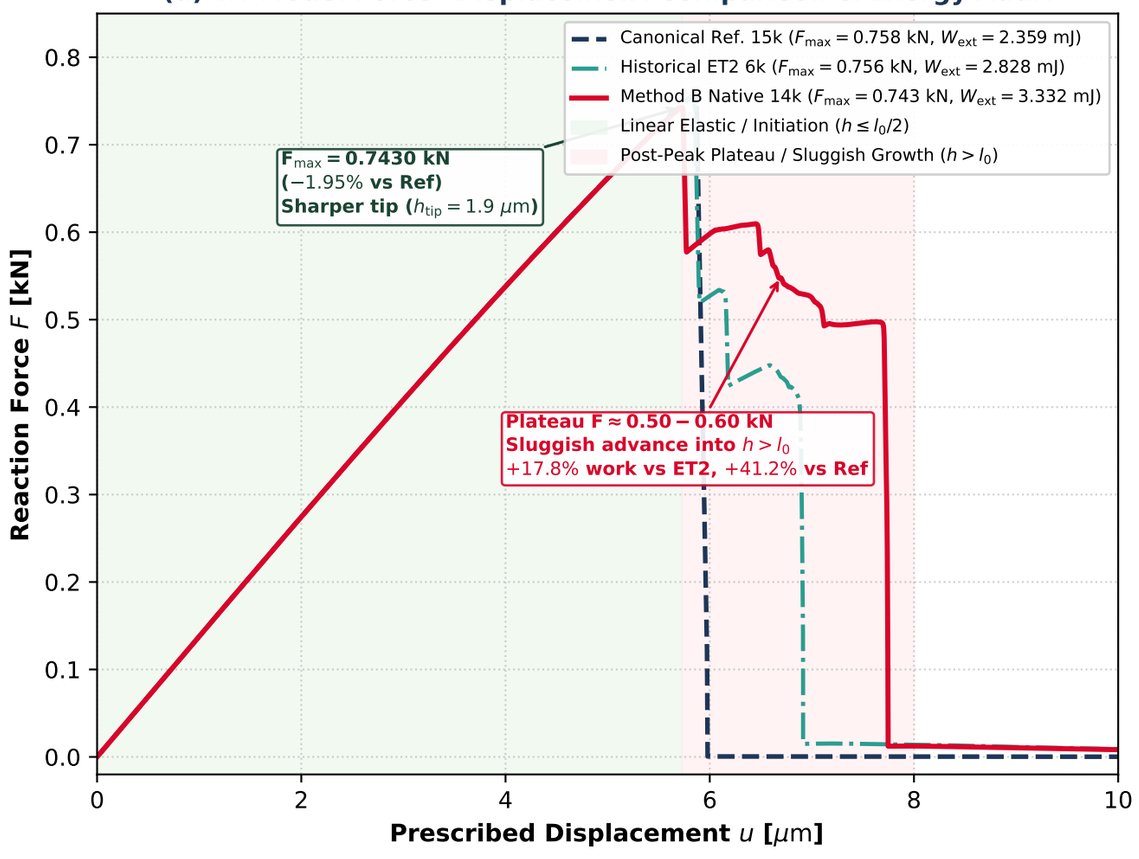

(a) Longitudinal Element Sizing Profile along Crack Ligament
Fine Initiation Zone ($h \le 3.75\ \mu m = l_0/2$)
Transition Corridor ($3.75 < h \le 7.5\ \mu m$)
Under-Resolved Ligament ($h > 7.5\ \mu m = l_0$)
Corridor Elements ($|y - 0.5| \le 25\ \mu m$)
10th--90th Percentile Band
Median Sizing $h(x)$
Length Scale $l_0 = 7.5\ \mu m$ (Under-resolution threshold)
Guideline $l_0/2 = 3.75\ \mu m$ (Adequate resolution)
$h_{tip} \approx 1.91\ \mu m$ ($l_0/h_{tip} \approx 3.91$)
$h \approx 10.5\ \mu m > l_0$
Coarse Plateau Region)
Characteristic Size $h = \sqrt{A}$ [μm]
Longitudinal Coordinate x [mm]


(b) Phase-Field Discretization Density along Forward Ligament
Median $l_0/h(x)$
Optimal Sizing ($l_0/h \ge 3$)
Adequate Guideline ($l_0/h = 2$, Reference)
Regularization Threshold ($l_0/h = 1$, Under-resolved if < 1)
$l_0/h_{tip} \approx 3.91$
(Initiation Well-Resolved)
$l_0/h \approx 0.70 < 1.0$
(Severe Under-Resolution)
Elements per Characteristic Length l_0/h
Longitudinal Coordinate x [mm]


(c) Crack Front Propagation across Discretization Zones
Pre-Peak: $u = 0.0050$ mm
Peak Load: $u = 0.0057$ mm ($F_{max} = 0.743$ kN)
Post-Peak: $u = 0.0059$ mm
Plateau 1: $u = 0.0065$ mm
Plateau 2: $u = 0.0070$ mm
Plateau 3: $u = 0.0075$ mm
Snap-Through: $u = 0.0080$ mm
Crack Front Threshold ($d \ge 0.95$)
Phase-Field Damage $d(x, y = 0.5\ mm)$
Longitudinal Coordinate x [mm]


(d) Tri-Model Force--Displacement Comparison & Energy Audit
Canonical Ref. 15k ($F_{max} = 0.758$ kN, $W_{ext} = 2.359$ mJ)
Historical ET2 6k ($F_{max} = 0.756$ kN, $W_{ext} = 2.828$ mJ)
Method B Native 14k ($F_{max} = 0.743$ kN, $W_{ext} = 3.332$ mJ)
Linear Elastic / Initiation ($h \le l_0/2$)
Post-Peak Plateau / Sluggish Growth ($h > l_0$)
$F_{max} = 0.7430$ kN
(−1.95% vs Ref)
Sharper tip ($h_{tip} = 1.9\ \mu m$)
Plateau $F \approx 0.50 - 0.60$ kN
Sluggish advance into $h > l_0$
+17.8% work vs ET2, +41.2% vs Ref
Reaction Force F [kN]
Prescribed Displacement u [μm]